

Incompressible Stokes Flow Annulus Benchmark Using Underworld3

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Abstract. We reproduce the annulus benchmark of Kramer et al. (2021) (hereafter K2021) using Underworld3 (hereafter UW3; Moresi et al., 2025). The benchmark considers incompressible, isoviscous Stokes flow in an annulus with analytical velocity and pressure solutions derived from prescribed buoyancy forcing. The benchmark includes both free-slip and zero-slip velocity boundary conditions, together with smooth and interface-localised density forcing. The UW3 calculations employ $P_2 \times P_1$ Taylor–Hood finite elements on progressively refined annulus meshes. Numerical accuracy is evaluated using relative volume L_2 -norm errors for velocity and pressure. For smooth forcing cases, the solutions recover approximately second-order pressure convergence together with systematic second-order or higher velocity convergence across the tested refinement range. Cases involving delta-function forcing exhibit reduced but consistent convergence behaviour, with velocity convergence rates approaching $O(h^{1.5})$ and pressure convergence rates approaching $O(h^{0.5})$, consistent with the reduced regularity associated with interface-localised buoyancy forcing. These results verify the UW3 annulus Stokes implementation for curved-shell geometries under both free-slip and zero-slip boundary conditions considered in K2021.

1 Introduction

Finite-element modelling has become an essential tool in geodynamics for investigating mantle convection and related solid-Earth processes in geometries that more closely represent planetary interiors. In particular, cylindrical annulus and spherical-shell domains are widely used because they naturally incorporate radial gravity and curved boundaries while avoiding some of the limitations of Cartesian approximations (e.g. Baumgardner, 1985; Jarvis, 1993; Zhong et al., 2000; Tackley, 2008; Davies et al., 2022). Accurate numerical treatment of incompressible Stokes flow in such geometries requires careful consideration of mixed finite-element stability, pressure nullspaces, curved boundary representation, and boundary-condition implementation. These issues become especially important when modelling free-slip boundaries and sharp density contrasts, both of which commonly arise in geodynamical applications.

Benchmarking and verification therefore play a central role in the development and validation of geodynamical modelling software. Community benchmark studies are widely used to assess numerical accuracy, convergence behaviour, and solver robustness across different finite-element discretisations and software frameworks (e.g. Blankenbach et al., 1989; Busse et al., 1994; King et al., 2010; Tosi et al., 2015; Thieulot and Puckett, 2018; Kramer et al., 2021; Thieulot and Bangerth, 2022). Benchmarks in curved annulus geometries are particularly valuable because they provide analytical solutions that enable di-

rect evaluation of velocity, pressure, and boundary-condition accuracy in geometries relevant to regional and global mantle dynamics.

In this study, we reproduce the annulus benchmark of K2021 using UW3. The benchmark considers incompressible isoviscous Stokes flow in an annulus with analytical velocity and pressure solutions derived from prescribed buoyancy forcing. The benchmark includes both free-slip and zero-slip velocity boundary conditions together with smooth and interface-localised density forcing. The smooth forcing cases assess convergence behaviour for regular analytical solutions, while the interface-localised forcing cases provide a more stringent test involving reduced solution regularity and sharp buoyancy variations across internal interfaces.

We investigate the convergence behaviour and numerical accuracy of the UW3 annulus Stokes implementation using $P_2 \times P_1$ Taylor–Hood finite elements on progressively refined annulus meshes. Numerical solutions are compared against the analytical benchmark solutions using relative L_2 -norm errors for velocity and pressure. Particular attention is given to the implementation of free-slip boundary conditions and the convergence behaviour associated with smooth and interface-localised forcing cases. The results provide verification of UW3 for curved-domain annulus Stokes flow and demonstrate the effectiveness of the K2021 benchmark for assessing finite-element accuracy in annulus geometries.

2 Governing Equations and Benchmark Solution

The benchmark solves the steady incompressible Stokes equations in an annulus $\Omega = \{(r, \theta) : R_- \leq r \leq R_+\}$:

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{g}, \quad (1a)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (1b)$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and $\mathbf{g}$ is gravitational acceleration. The stress tensor is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta\boldsymbol{\varepsilon}(\mathbf{u}), \quad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T), \quad (2)$$

with constant viscosity η .

Following K2021, the annulus radii are

$$R_- = 1.22, \quad R_+ = 2.22.$$

For the interface-localised cases, the internal density interface is located at

$$R_{\text{int}} = 2.0.$$

The analytical forcing is separated into azimuthal and radial parts. Smooth cases use a density perturbation proportional to

$$\left(\frac{r}{R_+}\right)^k \cos(n\theta), \quad (3)$$

where k controls the radial structure and n controls the azimuthal wavenumber. Delta-function cases use an interface forcing proportional to

$$\delta(r - R_{\text{int}}) \cos(n\theta). \quad (4)$$

The full analytical velocity and pressure fields are those derived by K2021; the UW3 implementation evaluates these closed-form solutions to define the forcing, impose the benchmark boundary conditions, and measure numerical error.

3 Numerical Setup

All models use the $P_2 \times P_1$ Taylor–Hood mixed finite-element pair, with quadratic velocity and continuous linear pressure discretisations. This element pair is widely used for incompressible Stokes flow because it satisfies the Ladyzhenskaya–Babuška–Brezzi stability condition and provides robust pressure behaviour for smooth solutions (Girault and Raviart, 1986; Boffi et al., 2013; Thieulot and Bangerth, 2022). The Stokes system is solved using the finite-element implementation in UW3, with the constant-pressure nullspace removed prior to error evaluation, since incompressible Stokes flow determines pressure only up to an arbitrary additive constant.

The benchmark suite consists of four case groups (Table 1) that systematically examine the effects of boundary conditions and buoyancy forcing. Cases 1 and 2 employ free-slip velocity boundary conditions, whereas Cases 3 and 4 use zero-slip boundaries. Cases 1 and 3 consider interface-localised delta-function forcing applied along an internal boundary, while Cases 2 and 4 use smooth volumetric forcing distributed throughout the domain.

Table 1. Benchmark cases used in the UW3 verification runs.

Case	Forcing	Boundary condition	Parameters
Case1	Delta function	Free slip	$n = 2, 8, 32$
Case2	Smooth	Free slip	$n = 2, 8, 32$; $k = 2, 8$
Case3	Delta function	Zero slip	$n = 2, 8, 32$
Case4	Smooth	Zero slip	$n = 2, 8, 32$; $k = 2, 8$

Figure 1 shows representative analytical density perturbations. Increasing n increases the angular complexity of the benchmark solution. For smooth forcing, increasing k changes the radial weighting of the density field. For delta-function forcing, the density is concentrated on the internal interface and therefore reduces solution regularity relative to the smooth cases.

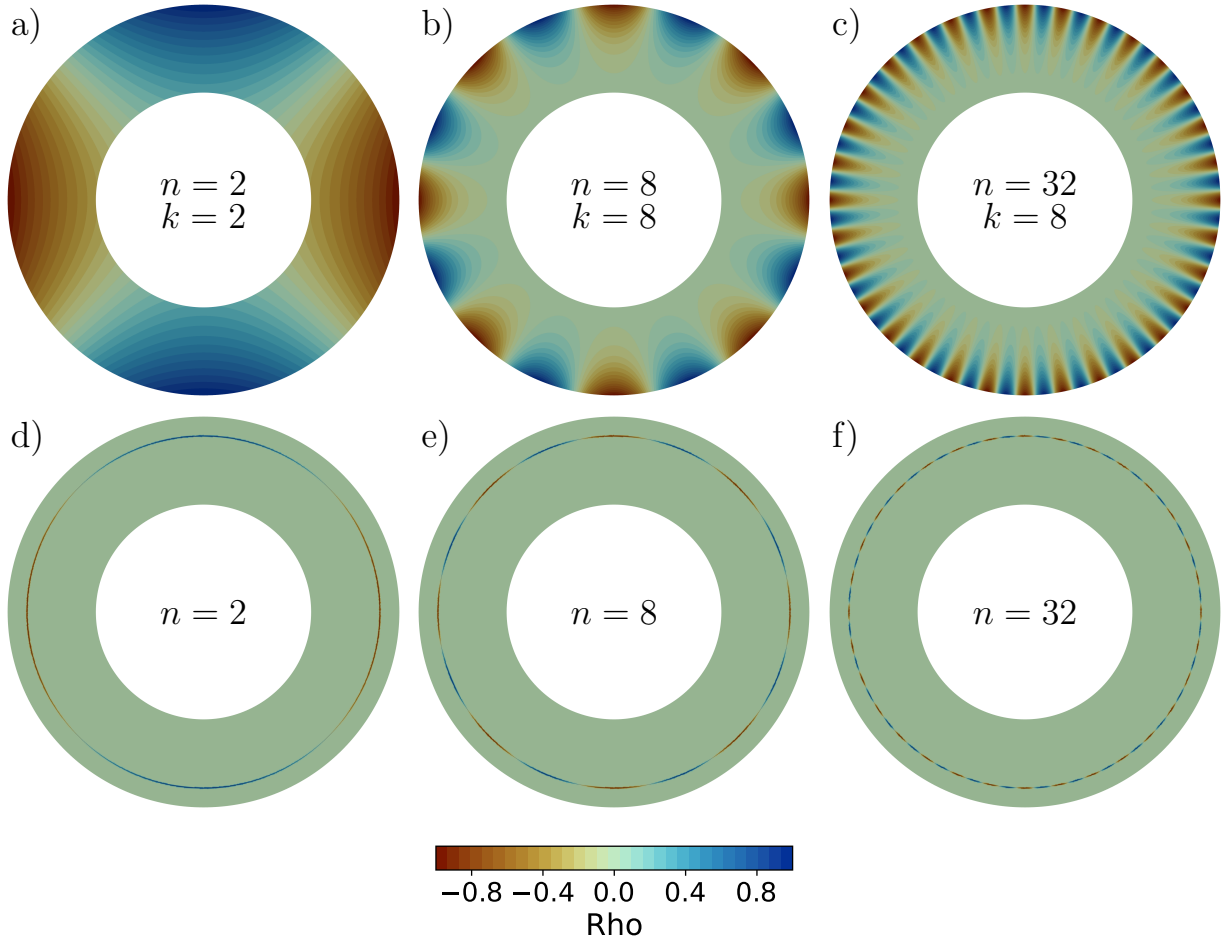


Figure 1. Analytical density perturbations. Panels a)–c) show smooth forcing and panels d)–f) show delta-function forcing. Columns correspond to increasing azimuthal complexity.

4 Error Metrics

Velocity and pressure errors are reported as relative volume L_2 norms. For a numerical field q_h and analytical reference field q^* , the relative error is

$$E_{L_2}^{\text{rel}}(q) = \left(\frac{\int_{\Omega} |q_h - q^*|^2 d\Omega}{\int_{\Omega} |q^*|^2 d\Omega} \right)^{1/2}. \quad (5)$$

Thus the table and figure labels use $E_{L_2}^{\text{rel}}(\mathbf{u})$ for velocity and $E_{L_2}^{\text{rel}}(p)$ for pressure. For pressure errors restricted to an annulus boundary Γ , we use

$$E_{L_2, \Gamma}^{\text{rel}}(p) = \left(\frac{\int_{\Gamma} |p_h - p^*|^2 d\Gamma}{\int_{\Gamma} |p^*|^2 d\Gamma} \right)^{1/2}, \quad (6)$$

where Γ denotes either the inner or outer boundary. For two consecutive refinements with characteristic mesh sizes h_1 and h_2 ,
80 the observed convergence rate is

$$\alpha = \frac{\log(E_{h_1}/E_{h_2})}{\log(h_1/h_2)}. \quad (7)$$

With uniform halving of the nominal mesh size, this reduces to $\alpha = \log_2(E_{h_1}/E_{h_2})$.

5 Results

The convergence results are shown in Fig. 2. For the smooth forcing cases, pressure errors converge close to $O(h^2)$ for both free-
85 slip and zero-slip boundary conditions. Velocity errors converge systematically and are approximately second order for $n = 2$
and $n = 8$. The high-wavenumber $n = 32$ smooth cases show larger error constants and stronger pre-asymptotic behaviour, but
the rates decrease toward the expected refined-mesh trend as the solution becomes better resolved.

The delta-function forcing cases exhibit lower rates. Velocity convergence approaches $O(h^{1.5})$, while pressure convergence
approaches $O(h^{0.5})$ for both free-slip and zero-slip boundaries. This behaviour is expected for the K2021 benchmark because
90 the Green's-function solution associated with interface-localised forcing is less regular than the smooth volumetric-forcing
solution. The reduced pressure rate is therefore interpreted as a property of the benchmark solution regularity rather than as a
failure of the Stokes solver.

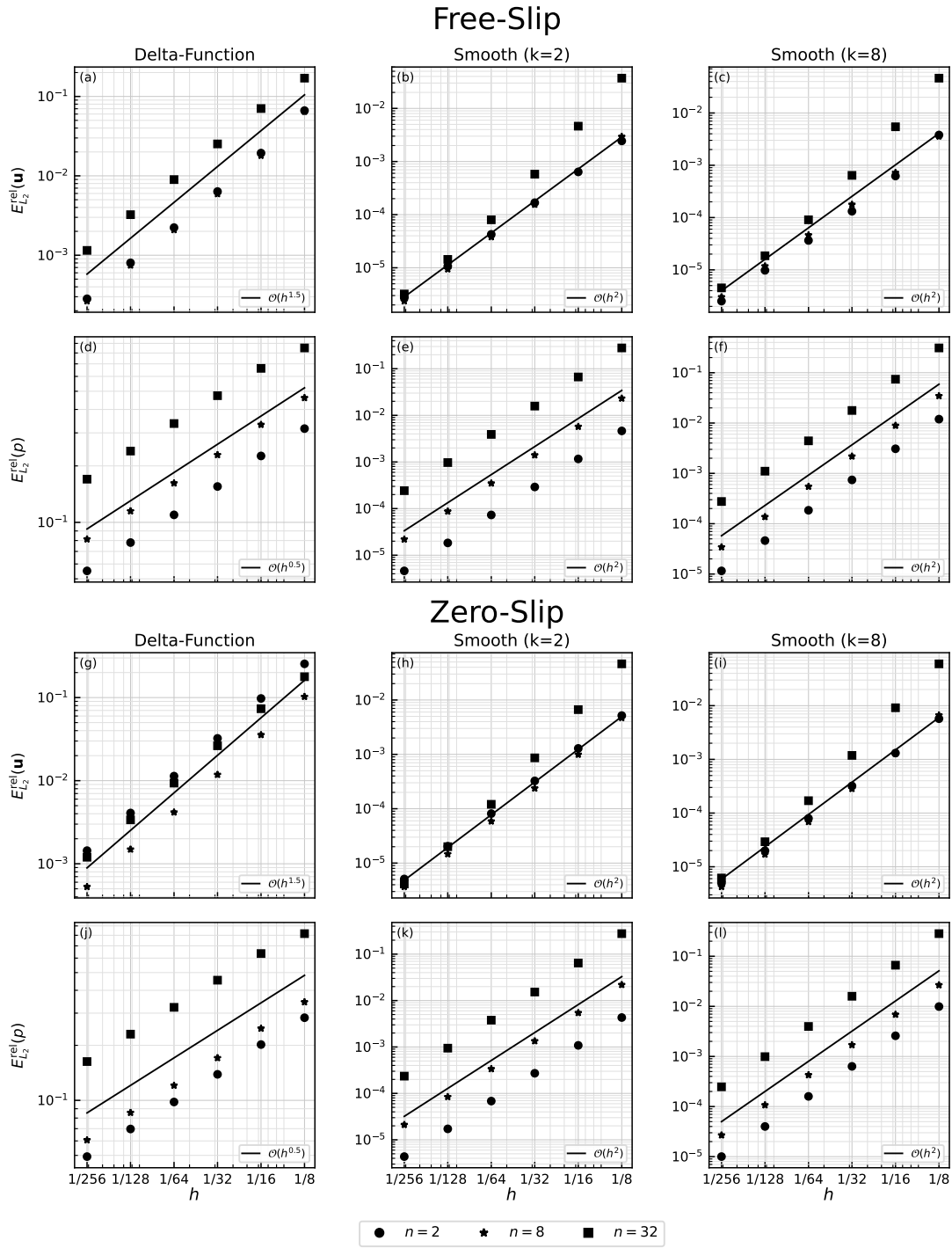


Figure 2. Velocity and pressure convergence. Columns show delta-function forcing and smooth forcing with $k = 2$ and $k = 8$. The upper block uses free-slip boundaries and the lower block uses zero-slip boundaries.

The convergence of pressure errors restricted to the inner and outer annulus boundaries is shown in Fig. 3. These boundary errors decrease close to the $O(h^2)$ reference trend for the smooth forcing cases. The delta-function cases also show approximately second-order boundary pressure convergence, even though the corresponding volume pressure errors converge more slowly. This distinction occurs because the volume norm includes the reduced regularity introduced at the internal forcing interface, whereas the boundary norm samples the pressure trace on the inner and outer boundaries. The inner and outer boundary errors have different constants, as expected for annulus geometry because the radii, boundary constraints, and distance from the internal interface differ.

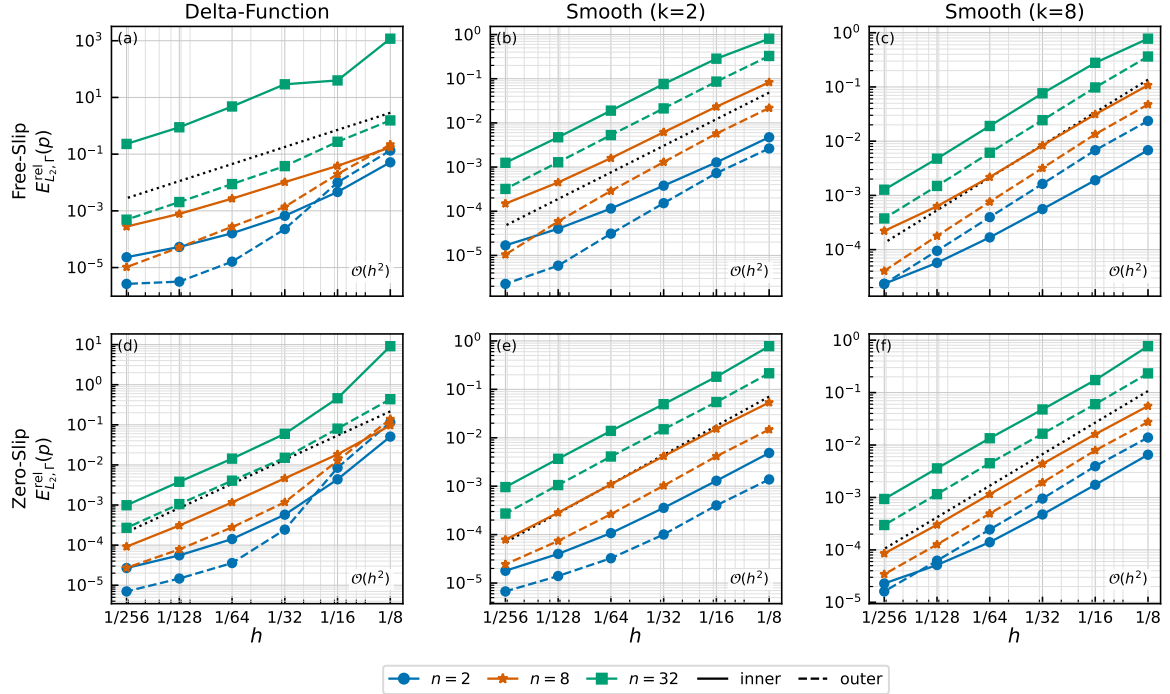


Figure 3. Convergence of relative boundary pressure errors $E_{L2,\Gamma}^{\text{rel}}(p)$. Solid curves denote the inner boundary and dashed curves denote the outer boundary. Colours indicate azimuthal wavenumber n .

The numerical errors and pairwise convergence rates are provided in Tables 2–7. Each table groups three azimuthal wavenumbers and reports mesh size, velocity error, velocity rate, pressure error, and pressure rate.

Table 2. Case1: Free-slip, delta-function forcing.

h	$n = 2$				$n = 8$				$n = 32$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	6.636e-02	–	3.161e-01	–	6.455e-02	–	4.610e-01	–	1.700e-01	–	8.511e-01	–
1/16	1.936e-02	1.78	2.260e-01	0.48	1.798e-02	1.84	3.318e-01	0.47	7.045e-02	1.27	6.624e-01	0.36
1/32	6.346e-03	1.61	1.553e-01	0.54	5.914e-03	1.60	2.290e-01	0.54	2.523e-02	1.48	4.736e-01	0.48
1/64	2.230e-03	1.51	1.097e-01	0.50	2.084e-03	1.50	1.617e-01	0.50	8.968e-03	1.49	3.364e-01	0.49
1/128	8.026e-04	1.47	7.804e-02	0.49	7.460e-04	1.48	1.149e-01	0.49	3.242e-03	1.47	2.400e-01	0.49
1/256	2.828e-04	1.50	5.512e-02	0.50	2.643e-04	1.50	8.132e-02	0.50	1.150e-03	1.50	1.699e-01	0.50

Table 3. Case2: Free-slip, smooth forcing ($k = 2$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	2.441e-03	–	4.639e-03	–	2.935e-03	–	2.313e-02	–	3.710e-02	–	2.796e-01	–
1/16	6.377e-04	1.94	1.164e-03	2.00	6.481e-04	2.18	5.728e-03	2.01	4.628e-03	3.00	6.660e-02	2.07
1/32	1.680e-04	1.92	2.911e-04	2.00	1.547e-04	2.07	1.401e-03	2.03	5.784e-04	3.00	1.580e-02	2.08
1/64	4.260e-05	1.98	7.312e-05	1.99	3.792e-05	2.03	3.487e-04	2.01	7.999e-05	2.85	3.904e-03	2.02
1/128	1.076e-05	1.99	1.836e-05	1.99	9.436e-06	2.01	8.722e-05	2.00	1.438e-05	2.48	9.718e-04	2.01
1/256	2.704e-06	1.99	4.611e-06	1.99	2.359e-06	2.00	2.189e-05	1.99	3.209e-06	2.16	2.425e-04	2.00

Table 4. Case2: Free-slip, smooth forcing ($k = 8$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	3.794e-03	–	1.195e-02	–	3.584e-03	–	3.454e-02	–	4.638e-02	–	3.106e-01	–
1/16	6.230e-04	2.61	3.080e-03	1.96	7.176e-04	2.32	8.903e-03	1.96	5.447e-03	3.09	7.427e-02	2.06
1/32	1.320e-04	2.24	7.416e-04	2.05	1.772e-04	2.02	2.180e-03	2.03	6.411e-04	3.09	1.777e-02	2.06
1/64	3.650e-05	1.85	1.848e-04	2.00	4.601e-05	1.95	5.461e-04	2.00	9.064e-05	2.82	4.427e-03	2.00
1/128	9.792e-06	1.90	4.617e-05	2.00	1.177e-05	1.97	1.367e-04	2.00	1.858e-05	2.29	1.105e-03	2.00
1/256	2.524e-06	1.96	1.154e-05	2.00	3.023e-06	1.96	3.424e-05	2.00	4.520e-06	2.04	2.761e-04	2.00

Table 5. Case3: Zero-slip, delta-function forcing.

h	$n = 2$				$n = 8$				$n = 32$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	2.545e-01	–	2.832e-01	–	1.025e-01	–	3.449e-01	–	1.783e-01	–	8.167e-01	–
1/16	9.755e-02	1.38	2.019e-01	0.49	3.555e-02	1.53	2.473e-01	0.48	7.344e-02	1.28	6.355e-01	0.36
1/32	3.228e-02	1.60	1.388e-01	0.54	1.181e-02	1.59	1.707e-01	0.54	2.629e-02	1.48	4.544e-01	0.48
1/64	1.134e-02	1.51	9.797e-02	0.50	4.163e-03	1.50	1.205e-01	0.50	9.343e-03	1.49	3.228e-01	0.49
1/128	4.083e-03	1.47	6.972e-02	0.49	1.490e-03	1.48	8.563e-02	0.49	3.378e-03	1.47	2.302e-01	0.49
1/256	1.439e-03	1.50	4.924e-02	0.50	5.281e-04	1.50	6.061e-02	0.50	1.198e-03	1.50	1.630e-01	0.50

Table 6. Case4: Zero-slip, smooth forcing ($k = 2$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	5.169e-03	–	4.309e-03	–	4.692e-03	–	2.188e-02	–	4.621e-02	–	2.766e-01	–
1/16	1.287e-03	2.01	1.083e-03	1.99	9.929e-04	2.24	5.433e-03	2.01	6.648e-03	2.80	6.437e-02	2.10
1/32	3.253e-04	1.98	2.715e-04	2.00	2.375e-04	2.06	1.343e-03	2.02	8.624e-04	2.95	1.527e-02	2.08
1/64	8.137e-05	2.00	6.831e-05	1.99	5.873e-05	2.02	3.358e-04	2.00	1.206e-04	2.84	3.784e-03	2.01
1/128	2.041e-05	2.00	1.721e-05	1.99	1.467e-05	2.00	8.419e-05	2.00	2.004e-05	2.59	9.432e-04	2.00
1/256	5.103e-06	2.00	4.351e-06	1.98	3.662e-06	2.00	2.111e-05	2.00	4.126e-06	2.28	2.354e-04	2.00

Table 7. Case4: Zero-slip, smooth forcing ($k = 8$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	5.720e-03	–	9.905e-03	–	6.657e-03	–	2.663e-02	–	6.009e-02	–	2.829e-01	–
1/16	1.311e-03	2.13	2.585e-03	1.94	1.283e-03	2.38	6.863e-03	1.96	9.136e-03	2.72	6.628e-02	2.09
1/32	3.202e-04	2.03	6.338e-04	2.03	2.837e-04	2.18	1.696e-03	2.02	1.185e-03	2.95	1.590e-02	2.06
1/64	7.962e-05	2.01	1.596e-04	1.99	6.861e-05	2.05	4.264e-04	1.99	1.699e-04	2.80	3.969e-03	2.00
1/128	1.990e-05	2.00	4.003e-05	2.00	1.700e-05	2.01	1.069e-04	2.00	2.923e-05	2.54	9.910e-04	2.00
1/256	4.970e-06	2.00	1.003e-05	2.00	4.234e-06	2.01	2.680e-05	2.00	6.181e-06	2.24	2.476e-04	2.00

6 Discussion

The convergence behaviour separates primarily by forcing type. Smooth forcing preserves the regularity needed for second-order pressure convergence with the $P_2 \times P_1$ discretisation. Velocity convergence is also regular, although the observed rates are
105 influenced by the curved geometry, boundary-condition enforcement, and the number of elements per azimuthal wavelength. These effects are most visible for $n = 32$, where the coarsest meshes under-resolve the angular structure.

The delta-function cases are more demanding because the forcing is concentrated on an internal circular interface. The slower convergence rates are consistent with the reference K2021 solutions and with the expectation that pressure is especially sensitive to reduced regularity in mixed Stokes discretisations. Importantly, the free-slip and zero-slip variants show the same
110 qualitative rates, indicating that the UW3 boundary-condition implementations behave consistently across both benchmark families.

The boundary pressure results provide a complementary diagnostic to the volume pressure error. They test how accurately the numerical pressure is represented where boundary tractions and normal stresses are evaluated. For the delta-function forcing cases, the boundary pressure can converge faster than the volume pressure because the singular forcing is located on an internal
115 interface rather than on the boundary itself. The boundary trace is therefore a smoother quantity than the full-domain pressure field measured in the volume norm. The different inner and outer boundary error constants should not be interpreted as an inconsistency: the two boundaries have different radii, different geometric quadrature paths, and different distances from the internal forcing interface. Agreement in the observed trends across free-slip and zero-slip cases is the more relevant check of the boundary-condition and pressure-evaluation implementation.

120 The benchmark is therefore a useful intermediate verification problem between Cartesian manufactured solutions and full spherical-shell mantle-convection calculations. It tests curved-domain Stokes assembly, pressure normalisation, boundary conditions, and both smooth and non-smooth buoyancy forcing while remaining compact enough for routine convergence testing.

7 Conclusions

We have presented summary of the UW3 Kramer annulus Stokes benchmark. The main findings are:

- 125 1. UW3 reproduces the analytical annulus solutions of K2021 for both free-slip and zero-slip velocity boundary conditions.
2. Smooth forcing cases show approximately second-order pressure convergence and systematic second-order or better velocity convergence over the tested mesh sequence.
3. Delta-function forcing cases show the expected reduced convergence, approximately $O(h^{1.5})$ in velocity and $O(h^{0.5})$ in pressure.
- 130 4. The observed distinction between smooth and interface-localised forcing reflects benchmark solution regularity and provides a useful test of finite element behaviour in curved geodynamic domains.

These results verify the UW3 implementation for annulus Stokes flow with analytical Kramer forcing and provide a compact reference for future annulus and spherical-shell benchmark comparisons.

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